in fact to have required a specific treatment. It is unfortunate that attractive extensions, such as correcting the Solow residual or introducing R & D, do not work very well. Clearly, the research program is at its beginning and models with more structural content are required. Such models would also be necessary to assess optimality properties. In the model we started with, recessions are caused by changes in preference and the question of the optimal number of recessions cannot arise. It might go differently in models with various causes for fluctuations where stabilization policies may face a trade-off between positive short-run effects and negative effects on growth.

References

Aghion, P. and G. Saint-Paul, 1991. On the virtues of bad times, CEPR Working paper no. 578 (CEPR, London).
Gali, J. and H. Hammour, 1991, Long-run effects of business cycles, Mimeo. (Columbia University, New York).

Comments

Productivity growth and the structure of the business cycle

by Gilles Saint-Paul

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This paper is an honest paper, it tackles an important question head on and confronts a model with the data. But it does not conceal from the reader the difficulties of matching the model to the data. The starting point of the paper is the observation that transitory disturbances to demand may have long-run effects on productivity growth. The author focuses on 'opportunity cost' models of productivity growth where the latter is measured in terms of the Solow residual. Certain activities which improve productivity, it is argued, have a comparative advantage during recessions. Investments with a more rapid pay-off are better carried out when demand is high, and investments with a longer-term pay-off have a comparative advantage when current demand is low.

The paper argues that the 'opportunity cost' model implies five predictions about the impact of demand shocks on productivity growth, and their correlations across countries. The discussion of these predictions and the interpretation of the results is admirably clear. But, as the author points out, most of these predictions are derived from the assumptions not only of the 'opportunity cost' model of productivity growth but also of an auxiliary assumption about the stochastic process generating demand shocks. It is, perhaps, not surprising therefore that the empirical results are not clear cut. The one robust result is that the short-run effect of a demand shock on productivity growth is negative. This is clearly supportive of the idea of the 'opportunity cost' structure of investment, but the results do not demonstrate conclusively that there is any path-dependence of output on demand shocks. Indeed, the relationship between the 'opportunity cost' view of investments and the existence of path-dependence of output is not straightfoward. It is possible to imagine that there exists a stock of innovations, the size of which is unaffected by demand shocks, but which are implemented at particular points of the cycle according to the comparative advantage implied by the model. In this case the impact of a given profile of demand shocks is merely to affect the timing of investment rather than the total level of investment or innovations that are made. The concept of comparative advantage of different types of investment does not, therefore, lead invevitably to path-dependence of output.

An important question raised in the paper is the question of how one distinguishes the trend in output from its cycle. This identification problem has two aspects, one theoretical and one empirical. At the theoretical level it is easiest to see the problem by considering a very simple example.

Consider a one-sector economy in which technical progress is both Harrod and Hicks neutral (i.e. Cobb–Douglas technology). Let A denote the level of technical knowledge. Suppose population is constant so that output per head is

$$y_t = A_t^{1-\alpha} \kappa_t^{\alpha}. \tag{1}$$

Denote capital per efficiency unit of labour by κ^*:

$$y_t = A_t \kappa_t^{*\alpha}. \tag{2}$$

Define

$$a_t = \ln A_t. \tag{3}$$

The equation for output becomes

$$\ln y_t = a_t + \alpha \ln \kappa_t^*. \tag{4}$$

The time-series process for output is non-stationary. The first term on the right-hand side is non-stationary (techical progress); the second term is stationary (adjustment toward a steady-state path). It is clear, therefore, that a possible decomposition of output is into these two terms. We could define the first term as the 'trend' and the second as the 'cycle'. This decomposition is not unique in the statistical sense. It does *not* correspond to the notion that the trend is the best forecast of output in the long run. Examples can be generated by taking specific forms for the determination of a_t. Two cases that we might study are (i) random walk for a stochastic exogenous technical progress, and (ii) endogenous growth models in which the growth rate of technical progress depends upon the state variables. In order to work with a stationary process it is natural to examine growth rates. Define the growth rate of output per head by

$$\eta_t = \ln y_{t+1} - \ln y_t. \tag{5}$$

This leads to the model

$$\eta_t = \Phi(\cdot) + \alpha(\ln \kappa^*_{t+1} - \ln \kappa^*_t), \tag{6}$$

where $\Phi(\cdot)$ is the 'technical progress function'. In models of exogenous growth it is simply a constant. In endogenous growth models it is a function of the state variable κ^* and the current shock s_t:

$$\eta_t = f(\kappa^*_t, s_t) + g(\kappa^*_t, s_t). \tag{7}$$

Clearly the trend and cycle are correlated and can be identified only by an identifying theoretical restriction. Typically $f(\cdot)$ and $g(\cdot)$ are non-linear.

As the author points out, the identifying restrictions appropriate to this sort of model are inconsistent with the restriction that was used by Blanchard and Quah in their semi-structural VAR estimation. Blanchard and Quah assume that the long-run effects of a demand shock on productivity are zero. Saint-Paul assumes that the immediate impact of a demand shock on productivity is zero. But only a more fully articulated model can produce the appropriate identifying restriction. In the case of a comparative advantage model of investments, but where the stock of innovations is given exogenously, then the Blanchard and Quah identifying assumption would remain appropriate. Saint-Paul's assumption is justified by the reasonable belief that productivity-improving activities undertaken in recessions take time to have an effect on productivity. But eq. (7) shows that in general the appropriate identifying restriction will vary according to the particular structural model assumed. And this viewpoint is taken up by Saint-Paul in his concluding comments.

On can only applaud, first, the ingenious correction to the observed Solow residuals, and, second, the reporting of the results with this corrected residual which are much less supportive of the model than those obtained with uncorrected residuals. And the author also points out fairly that his results could be interpreted within the framework of alternative models in which the trend and the cycle are correlated, such as the model analysed by Robson and myself. This model too produces path-dependence in output and the correlations that the author derives from his 'opportunity cost' model. The discussion of the results, and the attempt to interpret them in terms of a coherent theoretical framework, will be very helpful to those following in the author's footsteps.